

01 - Phenotype, Timing, and Endpoint

Companion reading handout for simultaneous listening.

Course: Medical Prevention of Spontaneous Early Delivery

Two patients can both have a cervical length of twenty-two millimeters, and yet the correct medical conversation may be different.

One patient is a singleton pregnancy, asymptomatic, with no prior spontaneous preterm birth, scanned at twenty-one weeks. The other patient has already delivered spontaneously at twenty-eight weeks in a previous pregnancy, and now the same cervical length is found later. The ultrasound number is identical. The clinical object is not identical.

That is where this course begins. Not with a drug. Not with a recommendation. With the discipline of naming the object of care.

The Three Questions

When we talk about medical prevention of early delivery, the first question is not, "Does short cervix get progesterone?" That question is too crude. The first question is: what phenotype is this, at what gestational age was it detected, and which endpoint are we trying to prevent?

Those three words should become the resident's first reflex: phenotype, timing, endpoint.

Phenotype means the patient population. Is she asymptomatic, or is she contracting? Is this a singleton or a twin pregnancy? Are membranes intact? Is there cervical dilation? Is there a previous spontaneous preterm birth? Has she already had two very early births? The same cervical length can live inside different phenotypes, and evidence belongs to phenotypes.

Timing means the gestational age at which the risk signal is found. A cervix found short at twenty-one weeks is not the same evidence situation as a cervix first found short at twenty-eight weeks. Timing is not decoration. Timing determines whether the patient stands inside the randomized-trial evidence window, near its edge, or outside it.

Endpoint means what outcome the study or guideline is trying to prevent. "Preterm birth" often means birth before thirty-seven weeks. But a birth at thirty-six plus five and a birth at twenty-seven plus five are not the same biological or neonatal event. Many trials therefore use earlier thresholds: before thirty-four weeks, before thirty-

three weeks, before thirty-two weeks, or before twenty-eight weeks. Other studies ask a different question: how many days were gained? What was the mean gestational age at delivery? Was neonatal morbidity reduced?

The phrase "early delivery" is therefore incomplete until we define it.

Cervical Length Is Not a Cliff

This matters because cervical length itself is not a cliff. The classic anchor is the Iams study from nineteen ninety-six. Cervical length was measured at twenty-four weeks, and the risk of spontaneous preterm delivery before thirty-five weeks rose progressively as the cervix shortened. At or below the tenth percentile, which was twenty-six millimeters, the relative risk was about six point two. At or below the fifth percentile, twenty-two millimeters, the relative risk was about nine point five. At or below the first percentile, thirteen millimeters, the relative risk was about fourteen.

The important memory is not just the numbers. The important memory is the shape of the relationship. The cervix behaved as a continuum of competence. The shorter the cervix, the higher the risk. That is why a cutoff such as twenty-five millimeters is useful, but it is not a biological cliff.

So how should the resident store twenty-five millimeters?

Store it as a risk and guideline boundary. It helps identify the short-cervix group. It does not mean that twenty-four millimeters is disease and twenty-six millimeters is health. The Iams data should protect you from that kind of false binary thinking.

Now add the next number: twenty millimeters.

Twenty millimeters is different. In Society for Maternal-Fetal Medicine Consult Series number seventy, for an asymptomatic singleton pregnancy without a prior spontaneous preterm birth, a transvaginal cervical length of twenty millimeters or less before twenty-four weeks is the stronger recommendation zone for vaginal progesterone. A cervical length of twenty-one to twenty-five millimeters is not ignored, but it becomes a shared decision-making zone.

So the memory is a pair, not a single number. Twenty-five names the risk signal. Twenty is where the recommendation becomes strongest in that specific asymptomatic singleton phenotype before twenty-four weeks.

Now listen to how endpoint changes the conversation.

Thirty-seven weeks is the broad preterm-birth boundary. It is useful for public health, birth statistics, and recurrence counseling. But many prevention trials care deeply about earlier thresholds because neonatal risk rises sharply as gestational age falls. Fonseca's trial used spontaneous delivery before thirty-four weeks as the primary outcome. The PREGNANT trial used preterm birth before thirty-three weeks. Meta-

analyses often report before thirty-four, before thirty-three, before thirty-two, and before twenty-eight weeks.

Do not let those thresholds blur together. Each one asks a different clinical question.

Before thirty-seven weeks asks: did the pregnancy reach term? Before thirty-four weeks asks: did we avoid early preterm birth, where neonatal morbidity becomes more serious? Before twenty-eight weeks asks: did we avoid the extreme-preterm zone, where each week is profoundly consequential?

This is also why time-to-delivery matters. A treatment may not convert every patient from preterm to term. But if it moves delivery from twenty-eight weeks to thirty-one weeks, that can be clinically meaningful even though the birth is still preterm by the under-thirty-seven definition. Conversely, a reduction in birth before thirty-seven weeks can be less impressive if it does not move severe neonatal outcomes.

Bedside Reflex

By the end of this course, you should be uncomfortable with an unqualified sentence like, "progesterone prevents preterm birth." You should want to ask: in whom, started when, by which route and dose, compared with what, and for which endpoint?

Return now to the two patients with a twenty-two millimeter cervix.

The first patient is asymptomatic, singleton, no prior spontaneous preterm birth, twenty-one weeks. She resembles the cleanest short-cervix progesterone evidence more closely. The number twenty-two millimeters places her in the short-cervix risk boundary, but not in the strongest twenty-millimeters-or-less recommendation zone. This is where shared decision-making can be meaningful, because the evidence gradient matters.

The second patient has a prior spontaneous preterm birth. Her twenty-two millimeters does not live in the same probability space. Her history changes baseline risk. It also brings other management categories into view, some of them mechanical and outside this course. For medical treatment, we must not reflexively substitute one progestogen for another. Prior preterm birth alone is not the same indication as short cervix, and vaginal progesterone is not a replacement for withdrawn intramuscular seventeen alpha hydroxyprogesterone caproate in a patient without a shortened cervix.

This is the foundation. Phenotype first. Timing second. Endpoint third. Treatment fourth.

If you keep that order, the rest of the evidence becomes much easier to remember because every number has a place to live.